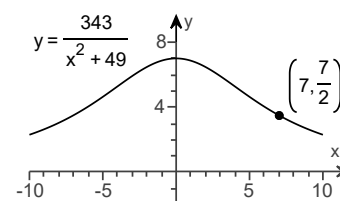


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Course: Calculus Beta

Assignment: MyLab Math - Uge 1

Find the tangent line to the Witch of Agnesi (graphed here) at the point $\left(7, \frac{7}{2}\right)$.



To find the tangent line to the curve $y = \frac{343}{x^2 + 49}$ at the point $\left(7, \frac{7}{2}\right)$, first find the slope of the curve.

To find the slope of the curve, use the Derivative Quotient Rule, which states if u and v are differentiable at x and if $v(x) \neq 0$ then the quotient

$$\frac{u}{v} \text{ is differentiable at } x, \text{ and } \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Find the slope of $y = \frac{343}{x^2 + 49}$. To do so, determine $\frac{dy}{dx}$ using the Quotient Rule.

$$\begin{aligned} \frac{d}{dx} \left(\frac{u}{v} \right) &= \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \\ \frac{dy}{dx} &= \frac{(x^2 + 49) \frac{d}{dx}(343) - 343 \frac{d}{dx}(x^2 + 49)}{(x^2 + 49)^2} \end{aligned}$$

Evaluate $\frac{d}{dx}(343)$ and $\frac{d}{dx}(x^2 + 49)$.

$$\begin{aligned} \frac{dy}{dx} &= \frac{(x^2 + 49) \frac{d}{dx}(343) - 343 \frac{d}{dx}(x^2 + 49)}{(x^2 + 49)^2} \\ &= \frac{(x^2 + 49)(0) - 343(2x)}{(x^2 + 49)^2} \end{aligned}$$

Simplify the numerator.

$$\begin{aligned} \frac{dy}{dx} &= \frac{(x^2 + 49)(0) - 343(2x)}{(x^2 + 49)^2} \\ &= \frac{-686x}{(x^2 + 49)^2} \end{aligned}$$

Let $y = f(x)$. Substitute 7 for x in the derivative and simplify to find the slope, $f'(7)$, of the line tangent at $x = 7$.

$$\begin{aligned} f'(7) &= \frac{-686(7)}{(7^2 + 49)^2} && \text{Substitute.} \\ &= -\frac{1}{2} && \text{Simplify.} \end{aligned}$$

Use the following formula for the tangent line at a point (x_0, y_0) , where x_0 and y_0 are real numbers and m is the slope at $x = x_0$.

$$y - y_0 = m(x - x_0)$$

Notice that this equation can be rewritten as $y = m(x - x_0) + y_0$. Substitute $-\frac{1}{2}$ for m , 7 for x_0 , and $\frac{7}{2}$ for y_0 and then simplify to find the equation of the tangent line at $x = 7$.

$$\begin{aligned}y &= -\frac{1}{2}(x-7) + \frac{7}{2} \\&= -\frac{1}{2}x + \frac{1}{2}(7) + \frac{7}{2} \\&= -\frac{1}{2}x + 7\end{aligned}$$

Multiply.

Simplify.

Therefore, the equation for the tangent line at $x = 7$ is $y = -\frac{1}{2}x + 7$.